Dynamic Planet C

Columbia University Science Olympiad Invitational

January 28, 2023

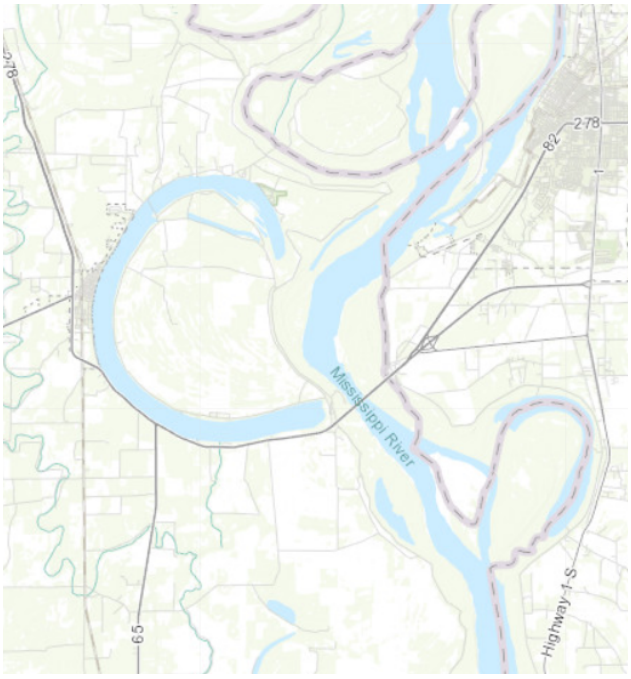


Figure 1: Thomas Cole, *The Oxbow* (1836). Currently located in the Met.

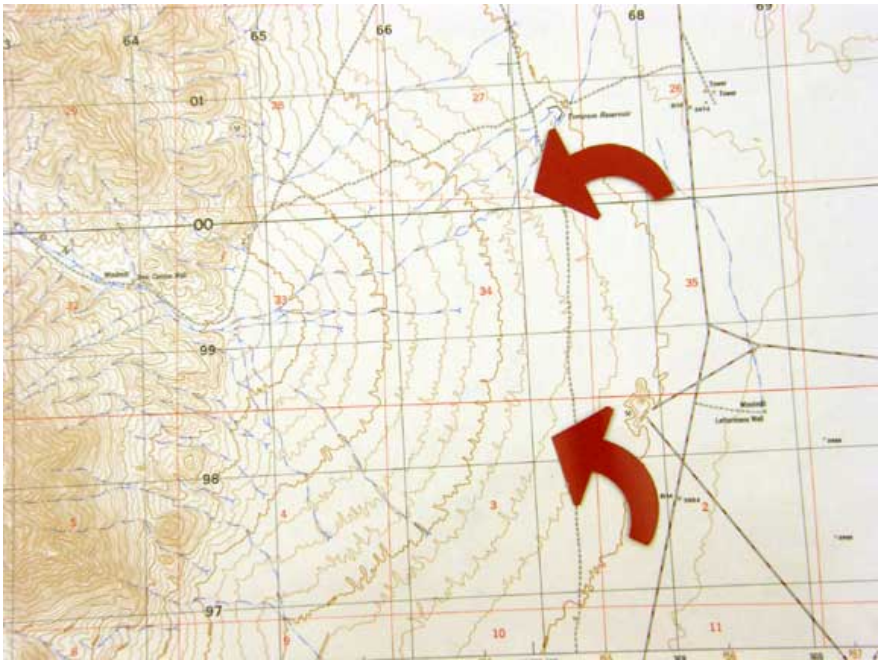
Please read all of the following bullet points.

- Do not open this test until you are told to begin.
- You have 50 minutes to complete the exam.
- This is the test packet for the exam. **Do not write on this document.** Please write only on the answer sheet and on any additional scratch paper.
- There are 72 questions for a total of 118 points.
- Good luck! :)

1 Topographic Map Interpretation



1. (1 point) What geological feature(s) are shown in the figure above? List up to two.

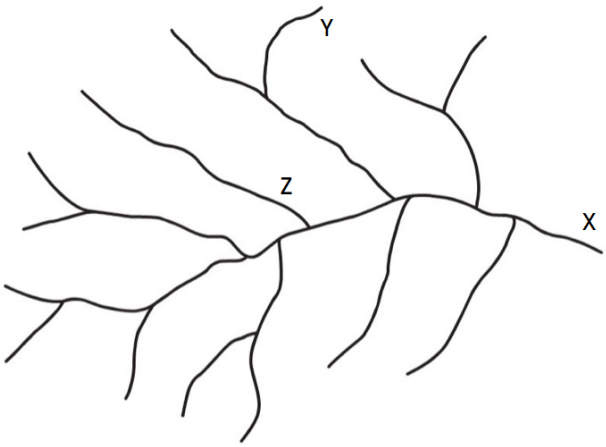


2. (1 point) What geological feature(s) are shown in the figure above? List up to two.



3. (1 point) What geological feature(s) are shown in the figure above? List up to two.

2 Rivers I



Refer to the diagram above for questions 4-9.

4. (1 point) What is the Shreve stream order at X?

5. (1 point) What is the Strahler stream order at X?

6. (1 point) What is the Hack stream order at X?

7. (1 point) X is an example of a(n) _____ stream.
   A. Consequent
   B. Subsequent
   C. Resequent
   D. Obsequent

8. (1 point) Y is an example of a(n) _____ stream.
   A. Consequent
   B. Subsequent
   C. Resequent
   D. Obsequent

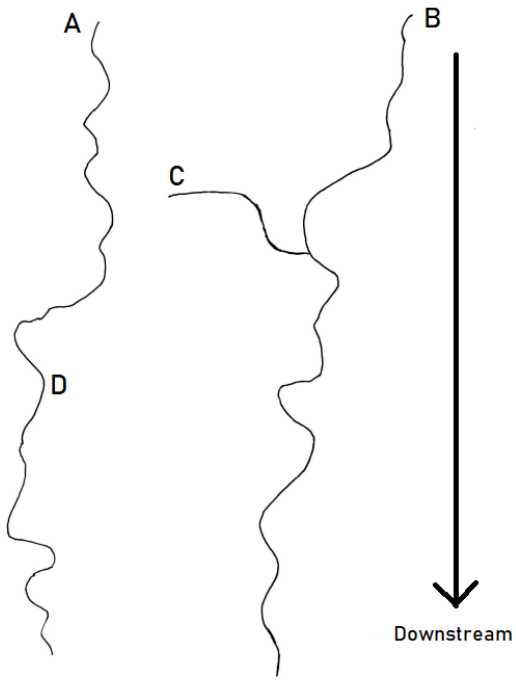
9. (1 point) Z is an example of a(n) _____ stream.
   A. Consequent
   B. Subsequent
   C. Resequent
   D. Obsequent

10. (2 points) Region A's bedrock consists of many anticlines and synclines of sedimentary rock with igneous intrusions. The bedrock has been eroded away over millions of years, exposing alternating layers of granite and shale on the surface. What drainage pattern does Region A most likely have and why?

11. (2 points) Region B consists of a dormant cinder cone volcano made primarily of basalt. What drainage pattern does Region B most likely have and why?

12. (1 point) Triple Divide Peak in Montana is a point where three North American drainage basins meet. How many continental divides meet at this point?

3 Rivers II



For the next 2 questions, refer to the diagram above depicting two streams with points A, B, C, and D.

13. (1 point) Assuming B is the main channel of the stream on the right, how did the tributary at C form?

14. (2 points) If the tributary at C continues to evolve, what will happen to the stream on the left? What will point D become?

15. (1 point) Where does the fastest moving flow of a meandering river occur?

    A. Along the point bar, near the river surface

    B. Along the point bar, near the riverbed

C. Along the cut bank, near the river surface

D. Along the cut bank, near the riverbed

16. (1 point) How does the formation of a meander affect the average gradient of a river?

A. Increase

B. Decrease

C. No Effect

17. (1 point) How would the formation of an oxbow lake affect the sinuosity of a river?

A. Increase

B. Decrease

C. No Effect

18. (1 point) True or false? All rivers drain into the ocean.

19. (1 point) A river has a true path length of 120 km and a straight line path length of 90 km. What is the river's sinuosity?

20. (2 points) What is the difference between an anastomosing stream and a braided stream?

21. (2 points) Some scientists hypothesize that the evolution of land plants coincided with an increase in the relative abundance of meandering rivers. What modern day evidence supports this hypothesis?

4 Sediments

22. (1 point) River sediments tend to become _____ the further downstream they are transported.

A. More angular, larger

B. More angular, smaller

C. Less angular, larger

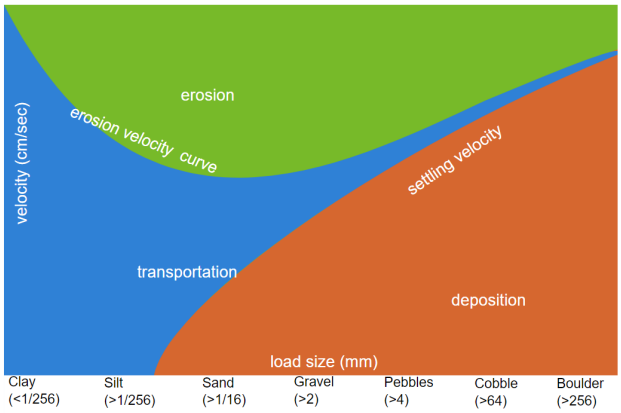
D. Less angular, smaller

23. (2 points) True or false? After a dam is built on a river, deposition will be enhanced downstream of the dam. Explain your answer.

24. (2 points) After a dam is built on a river, erosion will be enhanced upstream of the dam. Explain your answer.

25. (2 points) Suppose you have two rivers with equal discharge, the Silk River and the Tornado River. Due to differences in cross area and textures of their riverbeds, the Silk River has a more laminar flow, while the Tornado River has a more turbulent flow. Compare the Silk River and Tornado River in terms of competence and capacity. Which, if any, most likely has greater competence and why? Which, if any, most likely has greater capacity and why?

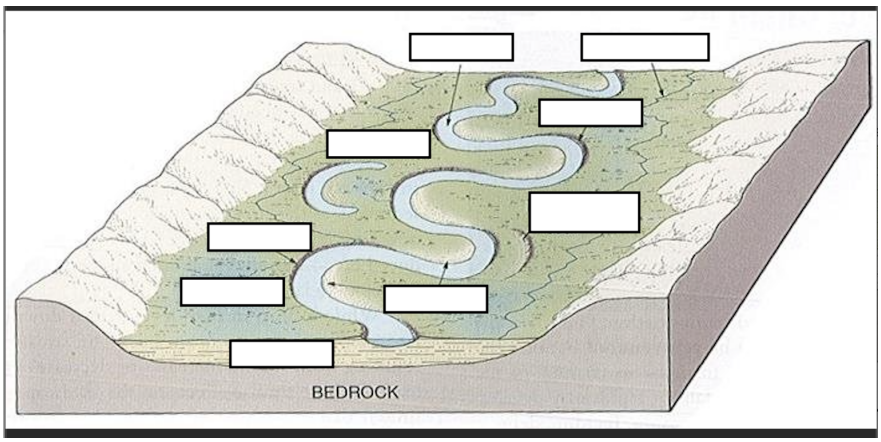
26. (1 point) How are potholes in a riverbed formed? Is the Silk River or the Tornado River more likely to have potholes?



27. Consider the Hjulstrom Diagram for streams (shown above). In this question we will explore the physical factors behind the characteristic shape of the curves in this diagram.
    (a) (2 points) Why is the erosion velocity higher than the settling velocity for all clast sizes? What term is used to refer to the sediment that travels at a velocity between the erosion velocity and settling velocity in a river?
    (b) (2 points) Why does the settling velocity go to zero for small clast sizes?
    (c) (2 points) Why does the erosion velocity sharply increase for small clast sizes?

5 River Valleys

28. (9 points) Label the features in the following floodplain diagram. Hint: One term will be repeated.



29. (2 points) Define a knick point. What do they signify?

30. (1 point) If a channel has potholes, it is likely to be a _____ channel.

    A. fluvial

    B. alluvial

    C. bedrock

    D. silt/clay

31. (1 point) What is the sinuosity of a straight, non-meandering river? Describe what occurs as sinuosity increases for a stream.

32. (1 point) A braided channel typically has which of the following material(s) as its primary composition? Select all that apply.

    A. silt

    B. gravel

    C. sand

    D. clay

33. (1 point) Why do natural levees take on the shape that they do?

6 Perennial streams & floods

34. (1 point) What does a low hydraulic radius indicate about water velocity?

35. (2 points) Can the final base level of a stream be lowered? If so, give an example of a situation in which it could be.

36. (3 points) Consider the forces and phenomena that govern the movement of water on Earth. One example would be gravity. Identify another significant phenomena that may play a role in determining the flow of a stream of water. Explain the role that it may play.

37. (2 points) What is an ephemeral stream? Where on Earth would you expect to find the greatest amount of ephemeral streams?

7 Groundwater

38. (2 points) Suppose that the water table lies above ground level for a river. How will the discharge of the river change as we go further downstream and why?

39. (2 points) Suppose instead that the potentiometric surface but not the water table lies above ground level for a river. How will the discharge of the river change as we go further downstream and why?

40. (2 points) In the context of streams and their water tables, there are three situations that may occur. The water table may be above the surface of the stream, below the base of the stream, or between the two. In which case(s) would there be a *losing stream*? In which case(s) would there be a *gaining stream*? Define the two terms.

41. (2 points) State a material that would be a good one for aquifers. Explain in terms of permeability and porosity.

8 Karst Hydrology

42. (1 point) Which rock cannot be commonly found in karst landscapes?

    A. Limestone

    B. Dolomite

    C. Sandstone

    D. Gypsum

43. (1 point) Using the associated letters (A, B, C, and D), rank the following descriptions of biomes from most likely to have karst topography to least likely.

A. Precipitation: 50 in/yr & Lots of flora
B. Precipitation: 10 in/yr & Little flora
C. Precipitation: 30 in/yr & Lots of flora
D. Precipitation: 30 in/yr & Little flora

44. (1 point) Speleothems in karst caves are most commonly made of
A. Chlorite
B. Magnesium
C. Sulfur
D. Calcite

9 Lakes I



45. (1 point) The depressions occupied by the lakes depicted in the figure above were primarily formed by _____.
A. Tectonic activity
B. Glacial activity
C. Fluvial activity
D. Volcanic activity

46. (1 point) What is the next greatest factor contributing to the formation of the lakes in the previous question?
A. Tectonic activity
B. Glacial activity
C. Fluvial activity
D. Volcanic activity

47. (2 points) Select all of the following ways that an endorheic lake could potentially lose water.

A. Evaporation

B. Fluvial drainage

C. Subsurface fluvial drainage

D. Percolation into groundwater

48. (2 points) True or false? Lakes can never be saltier than the ocean. Explain your answer.

49. (1 point) A small, freshwater, circular lake with no inlets or outlets is found in Minnesota. Approximately how long ago did this lake most likely form?

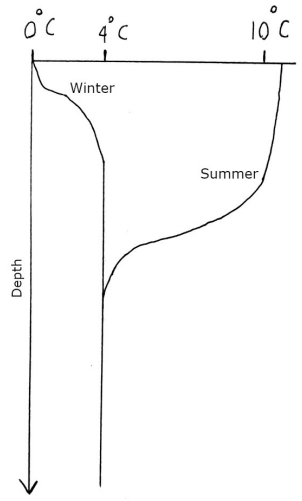
A. 1,000 - 2,000 years ago

B. 10,000 - 20,000 years ago

C. 100,000 - 200,000 years ago

D. 1 million - 10 million years ago

10 Lakes II



50. (1 point) Poisson Lake has different thermoclines during winter and summer as shown in the figure above. Based on the information given, what stratification does Poisson Lake have?

A. Cold Monomictic

B. Warm Monomictic

C. Meromictic

D. Dimictic

E. Polymictic

51. (1 point) During which month(s) does Poisson Lake most likely undergo turnover? Select all that apply.

    A. January

    B. April

    C. July

    D. October

    E. None of the above

52. (1 point) Which of the following layers reaches the coldest temperatures in most Antarctic lakes?

    A. Epilimnion

    B. Metalimnion

    C. Hypolimnion

    D. All have equal temperatures

53. (1 point) Which of the following types of lakes is most likely to have a halocline?

    A. Cold Monomictic

    B. Warm Monomictic

    C. Meromictic

    D. Dimictic

    E. Polymictic

54. (2 points) Compared to surface waves on lakes, internal waves _____ (Select all that apply)

    A. Have a greater density gradient

    B. Tend to have larger amplitudes

    C. Tend to have lower frequencies

    D. Are also driven by gravity

55. (3 points) Acid rain is a threat to aquatic ecosystems that can make lakes uninhabitable for fish and other organisms. Consider two lakes, Lake Alma and Lake Mater. Lake Alma is fed by a karstic aquifer system. Lake Mater is fed by a river system that cuts through a canyon made of sandstone. Which lake's ecosystem will be at less risk to harm from acid rain? Explain why and include any relevant chemical equations.

11 Wetlands and Destruction of Environment

56. (2 points) Due to groundwater over-extraction by the city of Allentown, local officials are concerned about the effects on nearby wetlands. Suppose that a bog and a fen are situated on opposite sides of the city, and are equally distant from the groundwater extraction site. Which one should the city be more concerned about and why?

57. (1 point) What is the difference between a marsh and a swamp?

    A. Marshes have lower pHs, while swamps have higher pHs

    B. Marshes have higher pHs, while swamps have lower pHs

    C. Marshes have herbaceous vegetation, while swamps have woody vegetation

    D. Marshes have woody vegetation, while swamps have herbaceous vegetation

58. (2 points) Select all of the following that could be an effect of constructing an artificial levee around a river.

    A. Decrease in species diversity of adjacent floodplain

    B. Greater facilitation of lateral nutrient exchange

    C. Reduction of aquifer recharge

    D. More frequent flood events

59. (1 point) Which of the following ground materials would experience the greatest amount of land subsidence from an equal amount of water being pumped out?

    A. Sand

    B. Gravel

    C. Clay

    D. Pebbles

60. (2 points) Why does your answer to the previous question undergo the greatest amount of land subsidence?

12 Hydrologic cycle

61. (4 points) The water balance equation can be written as follows:

$$P = Q + E + \Delta S$$

Identify each variable.

62. (3 points) Name three variables that can increase variable Q in a particular landscape.

63. (1 point) Which U.S state experiences the least rainfall yearly?

64. (2 points) Examine the following table detailing some data on various processes that occurred on Roareeland in May. How much infiltration did Roareeland experience?

Process	**Amount of Water** (km^3)
Transpiration	4
Runoff	13
Evaporation	6
Infiltration	7
Precipitation	22

13 Pollution

65. (1 point) Name the most radioactively polluted lake in the world.

66. (1 point) Describe one way in which water pollution may be reduced.

67. (1 point) Compared to an aquifer formed in a region of karst topography, would a standard aquifer be more able, equally able, or less able to purify polluted water?

14 Critical Zone Hydrology

68. (1 point) What does organic material do in a critical zone?

69. (1 point) What is a talik?

70. (2 points) What are pingos? What are the two types of pingos?

15 Darcy's Law

71. (2 points) Write out Darcy's Law. Identify any variables.

72. (1 point) Explain one limitation of Darcy's Law in regards to calculating groundwater flow.

Congrats! You made it to the end! Good luck with the rest of your events!